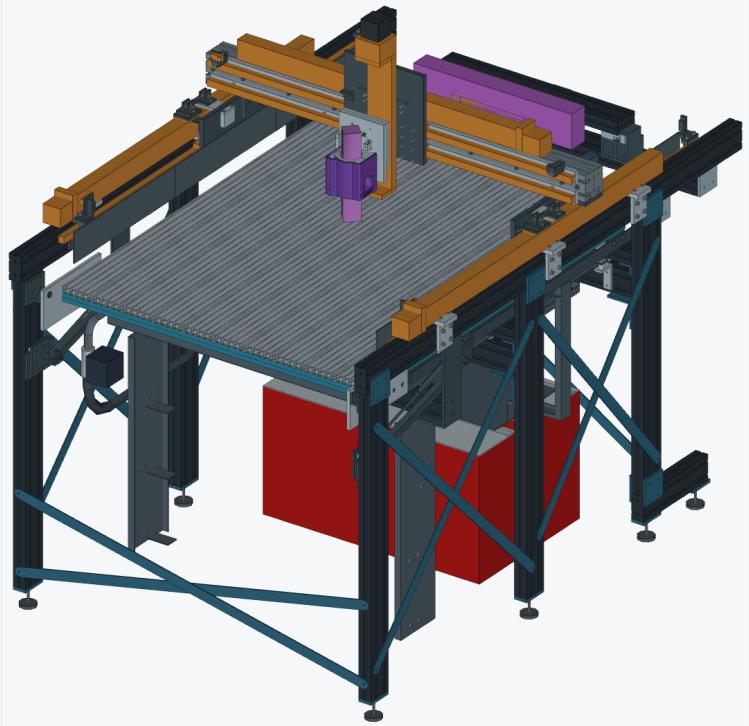
Integrated 4080 small table

Original owner head | Full HDPE / 20100 bed | Retractable ATC | 800 X / 1000 Y / 250 working Z

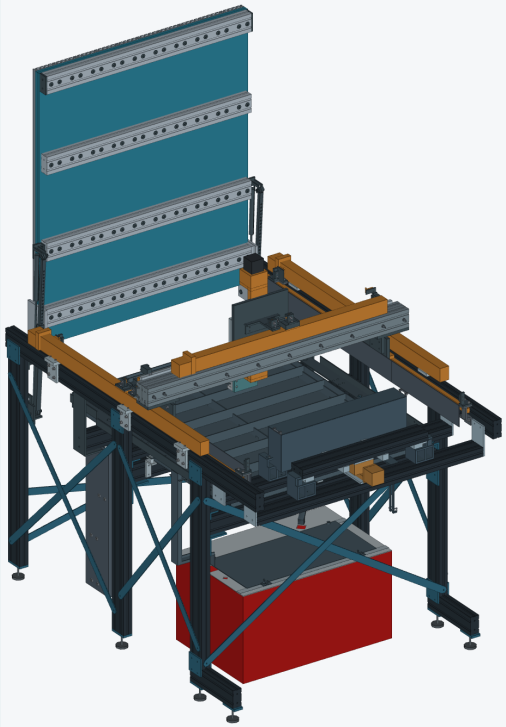
REVIEW MODEL - Budget and actual-interface requirements are not complete.

WOOD / CARVING



Ten uncut tracks, full drill-only HDPE layers and a removable billet head.

PLASMA / BED UPRIGHT



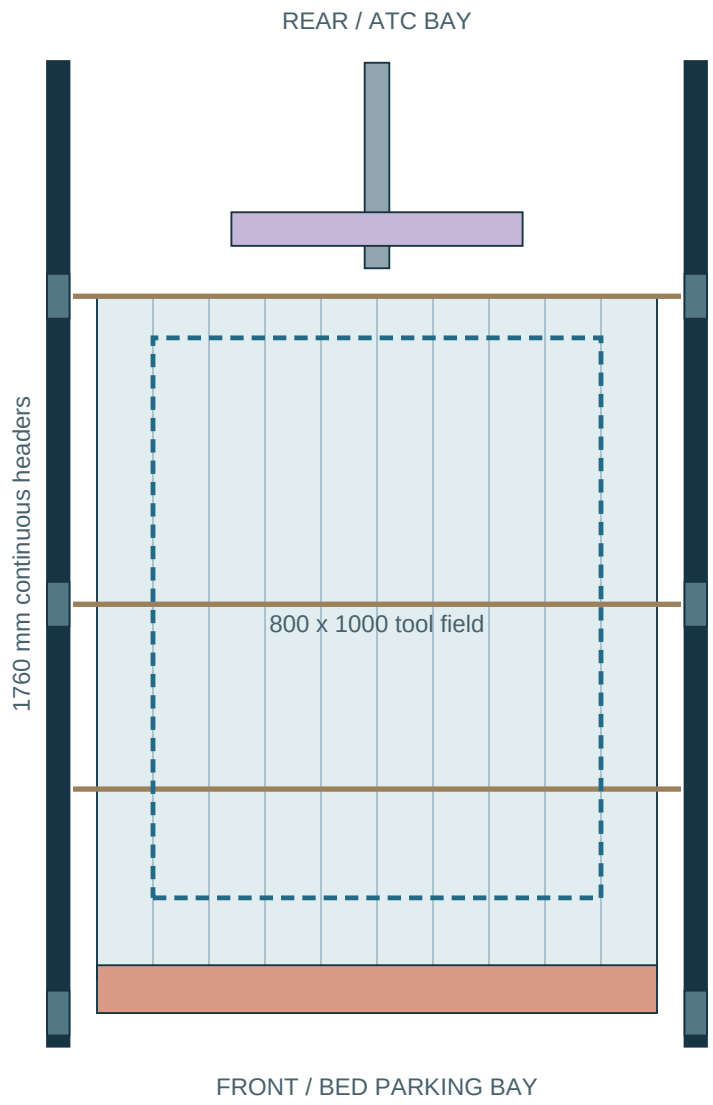
Bed parks in the front bay; covered magazine parks in the rear bay.

Rendered from the exported solid assemblies, including the original manufacturer magazine housing. Purple tool/spindle volumes are capacity context. The original owner boot remains at an unregistered reference pose.

The current source replaces earlier cassette beds and generic heads. It does not qualify the separate 4x8 design or steel machining.

Layout and load paths

Permanent front parking bay and rear ATC bay keep the bed and guard changes within the machine plan.



Defined geometry

Bed: 1000 x1220 mm; surface Z967.8. Ten uncut 1220 mm 20100 profiles over two 12.7 mm HDPE layers.

Hinge: Y-100, Z967.8. Upright bed Y-100 to -14.6; highest point approximately Z2247.8.

Frame: six posts and paired 1250 mm 4080 gantry beams. Pan supports at Y300 /630 /1180.

Head: original clamp/boot; removable billet dock plus one owned-stock side rib. Ideal hole-aware local bending: 0.093 mm at100 N; joints/torsion remain unqualified.

ATC: 200 mm slide; tool rows Y1100 /1300. Original manufacturer housing and pocket XY axes; slide-driven passive capture docks replace manual endpoint fasteners.

Front stops: cutting contact geometry is unchanged. Complete modules park 400 mm rearward, with guides stored and head removed, before bed tilt.

Analysis limits

The full-brim pan rib/floor screen is 1.156 mm without composite flange credit. It uses ideal supports and excludes weld distortion and thermal effects.

Member checks are not a full tool-to-work stiffness result. Actual sections, joint slip, guides, Z retention, vibration and safe handling need physical qualification.

All dimensional envelopes and continuous geometry proofs have explicit scopes. No firmware limits or machine settings were applied.

Cost and release status

Owned equipment is excluded. Unknown delivery charges and new hardware are not counted as zero.

Currently sourced required goods	USD, before unknown charges
Ten 1220 mm 20100 tracks - five two-profile packs	$399.95
One black 48 x96 x1/2-inch HDPE sheet - both layers	$225.75
RapidChange four-pocket ER11 Basic magazine	$400.00
Core goods extension	**$1,025.70**

Already $25.70 above the $1000 cap, before frame, pan/slats, custom stock, gas supports, reservoir/valve, hardware, tool setter, freight or tax. AliExpress hardware cart additions are separately recorded; they are not a full-build quote. Required five-pack track availability is not reserved.

Excluded owned equipment and stock

Motion modules, extra guides, drives/controller, spindle/VFD, existing working clamp/boot/actuator, electronics enclosure, cutter/torch, compressor and CV15HS are excluded from new equipment cost. The one half-inch and one three-eighth-inch aluminum squares have exclusive stock nests. No full-bed HDPE, tracks or RapidChange ownership credit is established.

Gates that prevent a final release

1. Register the working boot home/moving joint, spindle nose datum, selected ATC nut engagement, state sensing, received kit revision and exact vendor mating patterns.

2. Qualify received extrusion sections, slot nuts, modified members, joints, full rigidity, power-loss Z support and controlled bed handling.

3. Verify water capacity/temperature/overflow, actual strut and valve/fittings, module wipers/bellows, controller interlocks, VFD reverse/stop and isolated plasma control.

4. Resolve the new-parts delivered budget before ordering. Valid STEP solids and nominal clearance results do not close these requirements.

Current source, checks, stock and status: public project / variants / integrated-4080-small